Amazon Prime Video TV Shows & Movies

EDA and Machine Learning Insights Report

Project type: EDA + Supervised Classification

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At a glance

Metric	Value
Unique titles analysed	9,868 (8,511 movies, 1,357 shows)
Titles with an IMDb score	8,847 (used for modelling)
ML task	Predict IMDb score >= 7.0 from pre-release metadata
Selected model	HistGradientBoosting (tuned)
Test F1 / Recall / Precision	0.644 / 0.812 / 0.534
Test PR-AUC / ROC-AUC	0.648 (chance 0.249) / 0.854

1. Business problem and data

Amazon Prime Video keeps expanding its library, but it is hard to know what the catalogue holds, how it has changed and what audiences rate highly. The analysis covers four questions: content diversity, regional availability, trends over time, and what is associated with a high IMDb rating. Two datasets were used: **titles** (9,871 rows, 15 columns) and **credits** (124,235 rows, 5 columns), joined on **id**.

Data quality and preparation

- 3 duplicated title records removed (9,868 unique ids); duplicate credit rows removed.
- Credits are one-to-many, so they were aggregated to one row per title (actor and director counts) before merging; 1,007 titles have no credits.
- Missing values: **seasons** set to 0 for movies (structural), **age_certification** set to 'Unrated' (65.7% missing). Ratings were deliberately **not imputed** (imdb_score is missing for 1,021 titles).
- Derived features: genre and country lists, n_genres, n_countries, n_actors, n_directors, decade; IQR outlier flags for runtime, votes and popularity.

2. EDA insights

Fifteen charts followed the univariate, bivariate and multivariate structure. The main findings:

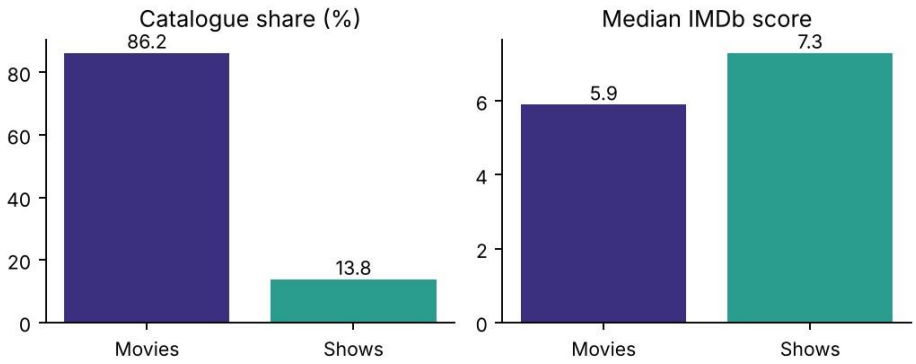
Content diversity and regional mix

- **Movie-heavy catalogue:** movies are 86.2% (8,511) and shows 13.8% (1,357).
- **Genres:** 19 genres, 2.24 per title on average. Drama (4,762), comedy (2,987) and thriller (2,119) lead; family (751) and animation (475) are thin.
- **Countries:** the US has 5,331 titles (58.9% of titles with a country listed), then India (1,072), Great Britain (928), Canada (530) and France (250). The top 5 make up 79.1% of country mentions across 117 countries.
- **Age certification:** only 34.3% of titles have one. Among rated titles R leads (36.9%), and children's TV ratings are small (about 5.5% combined).

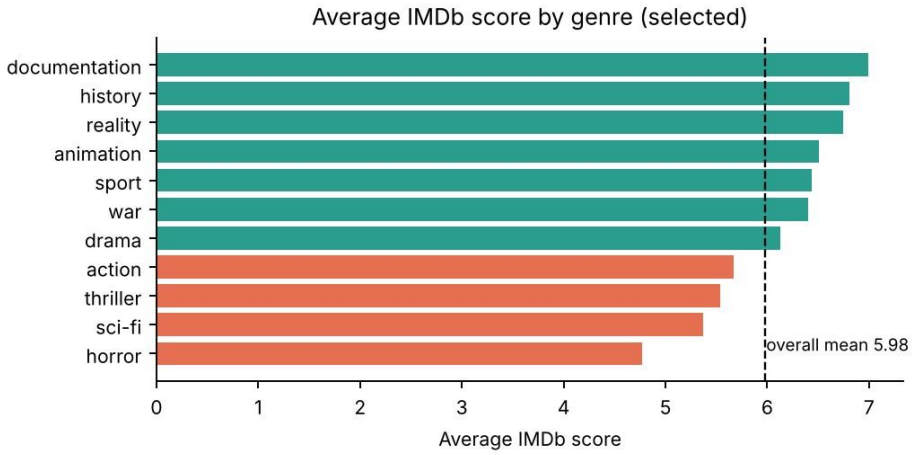
Trends over time

- Output is low and flat for decades, then climbs after 2000: 72.4% of titles were released since 2000 and only 17.9% before 1980.
- The peak is 2021 with 856 titles. The fall in 2022 (89 titles) most likely reflects a partial year of data, not a real decline.

Ratings and popularity



- IMDb scores average 5.98 (median 6.10). Only 6.1% of titles score 8.0 or above and 21.0% score below 5.0.
- **Shows beat movies:** median 7.3 vs 5.9. Shows score higher in every cast-size bucket too.
- **Best genres:** documentation 6.99, history 6.81, reality 6.75. **Weakest:** horror 4.77, sci-fi 5.37, thriller 5.54, action 5.67.
- **Decades:** averages stay in a narrow band (about 5.8 to 6.1); no clear long-term trend. The 2020s (5.77) are a partial decade.
- **Votes vs score:** Spearman 0.143, a weak link, so popularity is a poor proxy for quality. Cast size is negligible (-0.035).
- **Runtime:** movies average 94.0 min; shows average 35.8 min per episode, so the two are not like for like.



Correlation findings

IMDb score correlates most with **tmdb_score (0.647)**, then seasons (0.336) and imdb_votes (0.209). Release year, n_genres and runtime show almost no linear relationship. Because tmdb_score is another audience rating of the same title, it was excluded from the model to avoid target leakage.

3. Hypothesis tests (alpha = 0.05)

#	Hypothesis	Test	Result
1	IMDb score differs between movies and shows	Mann-Whitney U, two-sided	p = 9.5e-227; medians 5.9 vs 7.3. Reject H0
2	IMDb votes and score are positively related	Spearman, one-sided (n = 8,837)	rho = 0.143, p = 3.9e-42. Reject H0 (weak, about 2% of rank variation)
3	Documentaries score higher than other titles	Mann-Whitney U, one-sided	p = 4.1e-153; medians 7.1 vs 5.9; rank-biserial 0.522. Reject H0

All three nulls were rejected. The large samples make even small effects significant, so effect sizes matter: type and documentary status show large effects, votes a small one.

4. Machine learning

Problem and setup

Binary classification: is a title highly rated (IMDb score >= 7.0), using only metadata known before audience ratings exist. The threshold was chosen from the actual class balance:

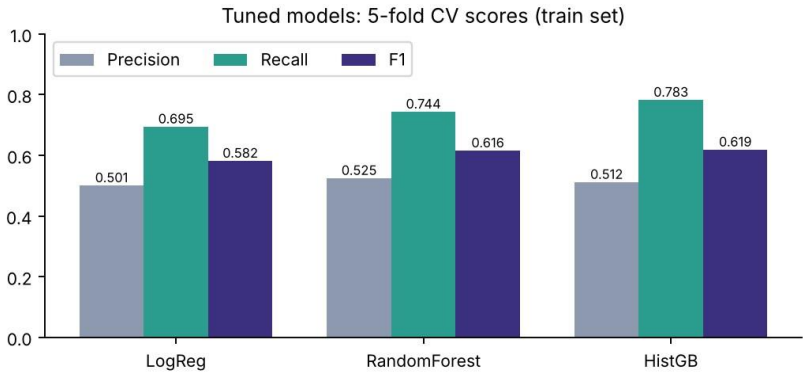
Threshold	Share of scored titles that are positive

6.5	37.8%
7.0 (chosen)	24.9% (about 3:1 imbalance)
7.5	13.2%

- **Rows:** 8,847 scored titles; unscored titles dropped, not imputed. Stratified 80/20 split: 7,077 train, 1,770 test (24.9% positive in both).
- **Features (38):** type, release_year, runtime (capped at 240 min), seasons, age_certification, n_genres, n_countries, n_actors, n_directors, 19 multi-hot genres, 10 top-country flags.
- **Excluded to avoid leakage:** tmdb_score, imdb_votes, tmdb_popularity, ids, title, description, top_cast, director, outlier flags.
- **Pipeline:** imputation, scaling and one-hot encoding sit inside scikit-learn Pipelines, so they fit on training data only.
- **Imbalance (about 3:1):** class_weight='balanced' in every model. SMOTE was not needed because recall was already adequate.

Model comparison (5-fold stratified CV, RandomizedSearchCV scored on F1)

Model	Precision	Recall	F1	PR-AUC	Best parameters
Logistic Regression	0.501	0.695	0.582	0.572	C = 0.0746
Random Forest	0.525	0.744	0.616	0.627	400 trees, depth 10, min leaf 5, sqrt features
HistGradientBoosting	0.512	0.783	**0.619**	0.603	leaf 100, 31 leaf nodes, depth 5, lr 0.1, L2 1.0

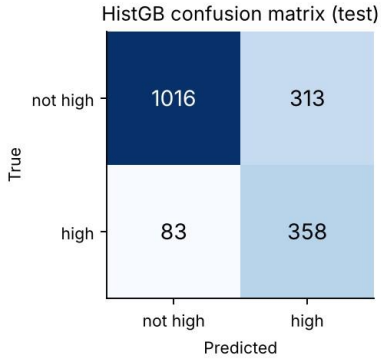


- Tuning barely helped Logistic Regression (F1 0.580 to 0.582), helped Random Forest a lot (0.527 to 0.616, recall 0.451 to about 0.74) and helped HistGB slightly (0.609 to 0.619).
- HistGB was selected on the highest CV F1 (it is close to Random Forest) and the highest CV recall.

Final test-set results (evaluated once)

Class	Precision	Recall	F1	Support
not high	0.92	0.76	0.84	1,329
high	0.53	0.81	0.64	441
Accuracy			0.78	1,770

Test F1 0.644, PR-AUC 0.648 (chance level 0.249), ROC-AUC 0.854. The test F1 is in line with the CV F1 (0.619), so there is no sign of overfitting.



What drives the predictions

Permutation importance ranks **seasons**, **documentary genre**, **runtime**, **number of actors** and **release year** as the top five, followed by horror, India as a production country, thriller and action. Seasons and runtime largely separate shows from movies, which matches the EDA. Importance shows what the model relies on, not what causes a high rating.

5. Business impact and recommendations

What the metrics mean

- **Recall 0.812:** the model finds about 81% of truly highly rated titles (358 of 441); 83 are missed.
- **Precision 0.534:** about 47% of flagged titles are false alarms (313 of 671 flagged), which means wasted spend if used alone.
- **Use:** a first-pass screening tool to shortlist candidates for review. Raising the decision threshold would trade recall for precision.

Recommendations

1. **Expand TV series.** Shows are 13.8% of titles but have a median score of 7.3 vs 5.9 for movies.
2. **Invest in strong genres.** Documentary, history and reality rate highest; family and animation are thin and would serve households with children.
3. **Curate weak, high-volume genres.** Horror, sci-fi and thriller rate below the 5.98 average; acquire selectively and review low-rated titles.
4. **Broaden regional content.** The US is 58.9% of titles with a country; India is a clear second. More Indian, Japanese, German and Italian content reduces US dependence.
5. **Fix metadata gaps.** 65.7% of titles lack an age certification and about 10% lack an IMDb score, which hurts parental controls and recommendations.
6. **Do not rely on cast size, release decade or votes alone;** none explains ratings well.
7. **Use the model for screening,** not as the only decision-maker.

6. Limitations and next steps

- Features are metadata only; precision is moderate (0.534).
- Ratings are associations, not causes, and rated titles may not represent all titles.
- Next steps: add description text (e.g. TF-IDF) or cast history as features, and tune the decision threshold to the business cost of false alarms vs missed titles.

- The final pipeline was saved with joblib and reloaded for a sanity check.

All figures in this report come from the executed notebook outputs. The permutation importance ranking is read from the chart, so no exact importance values are quoted.